

What is quantum resonance?

— resurgence theory —

Okuto Morikawa and Shoya Ogawa

(nuclear)
arXiv: 2503.18741

2505.02301

2508.09211

2510.11766

② Quantum Theory

$$H\psi = E\psi$$

I. harmonic oscillator



$$E_n = \hbar \left(n + \frac{1}{2} \right)$$

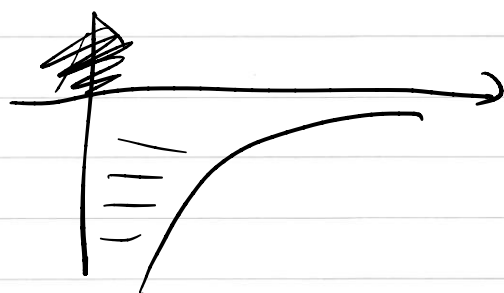
$\Rightarrow$ eigenvalue problem.

H : Hermitian op.
 $\therefore E$: real

$$H = H(x, p), [x, p] = i\hbar$$

in particular, $p = -i\hbar \frac{d}{dx}$

2. Coulomb



} continuous

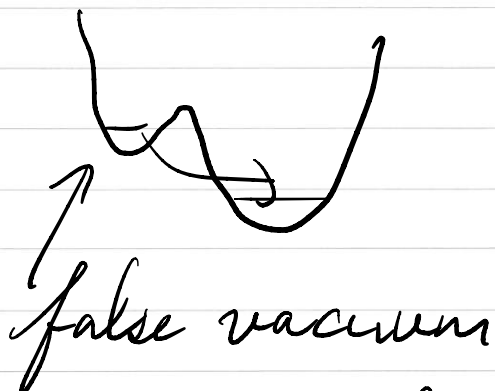
} eigenvalue

k $\rightarrow$ cross-section
(given k)

$\Rightarrow$ scattering problem
in general,



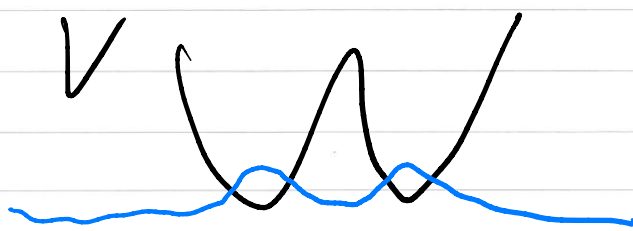
3. Quasi-stationary



quantum tunneling
non-perturbative

usually, quantum theory
is in bounded potential V

→ at infinity (asymptotically)
 ψ should damp.

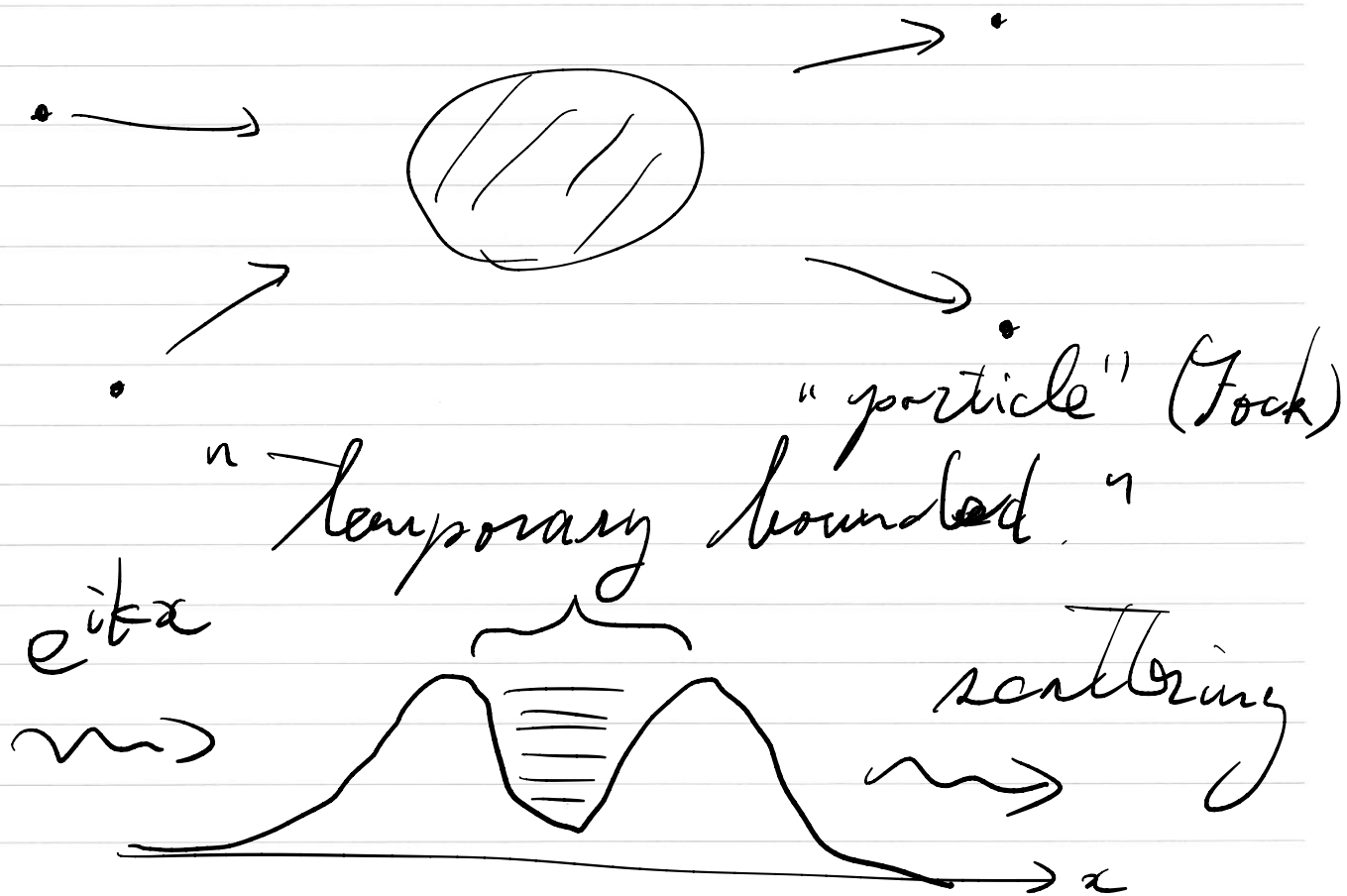


ψ ground state
and eigenvalue is well-defined.

However, perturbation theory
choose one vacuum,
which is unstable with
complex energies

②

Resonance
asymptotically free $V(r) \sim 0$
in QM, QFT.



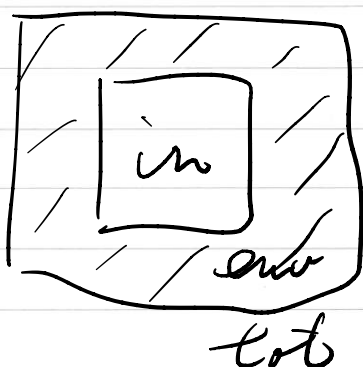
Transmission has peaks
at these discrete energies
complex due to decay

1. nuclear reaction

heavy particle $\sim$ decay

2. non-Hermitian, open system

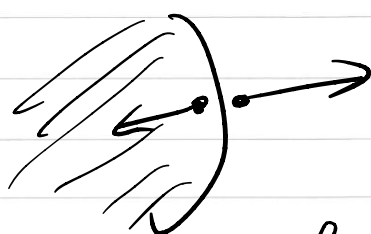
After integrated out "env",



$$H_{\text{tot}} \xrightarrow{\text{env}} H_{\text{in}}$$

H_{in} is not Hermitian

3. Black hole



quasi-normal mode

event horizon

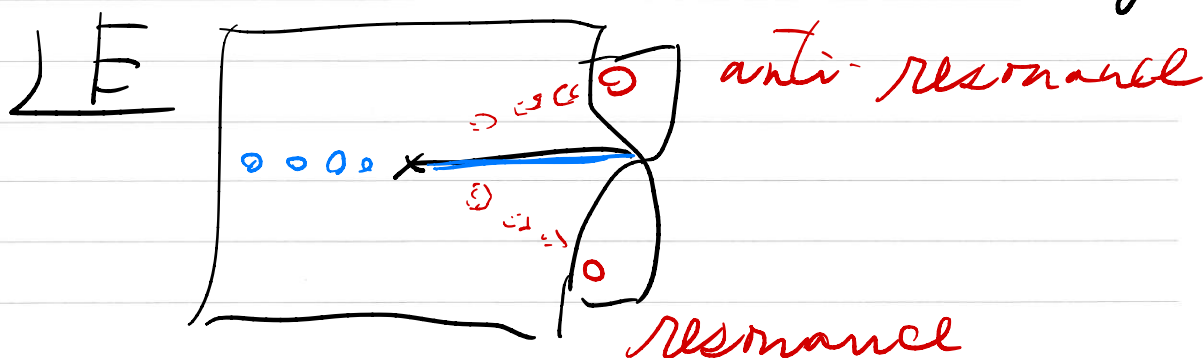
J. (Miyachi et al. '25)

② How to see resonance typically,



1. S -matrix (Transmission)

$S[k]$ has pole singularity.



2. resolvent

$$F[\psi](z) = \langle \psi | \frac{1}{H - z} | \psi \rangle \quad z \in \mathbb{R}$$

$\leadsto$ analytic continuation

at $z = E_{\text{res}}$ ill-defined $z \in \mathbb{C}$

3. regularized state

some techniques
(phenomenological)

② Complexification

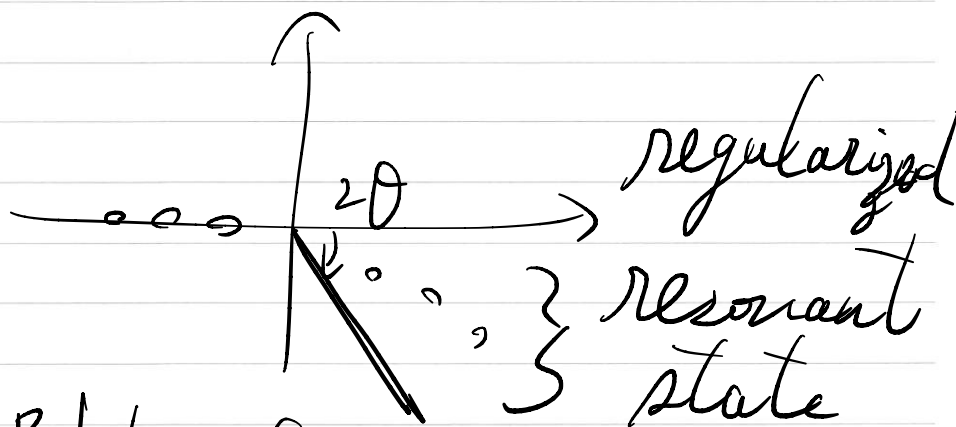
naively, ψ_{res} is ill-defined

$$\psi \xrightarrow{x \rightarrow \infty} e^{ikx} = e^{i(k_r x + k_i x)} \sim \infty$$

1. complex scaling

$$x \rightarrow x e^{i\theta}$$

$\underline{LE}$



Aguilar-Balslev-Combes $\int \|\psi\|^2 < \infty$
Theorem [71]

resonant energy = eigenvalue

2. resurgence $\leftarrow$

Where is an integrable line
on $x \in \mathbb{C}$?

perturbation theory suffers from
divergent asymptotic series.

$$f = \sum_n \hbar^n f_n = \infty$$

$$f_n = \frac{n!}{n^n}$$

WKB

$$\psi = e^{iS/\hbar}$$

Resum
over $\hbar \in \mathbb{C}$
with $x \in \mathbb{C}$

$$S \sim \pm \frac{1}{\hbar} \int \sqrt{V-E}$$

$$\psi = \boxed{\text{divergent}}$$

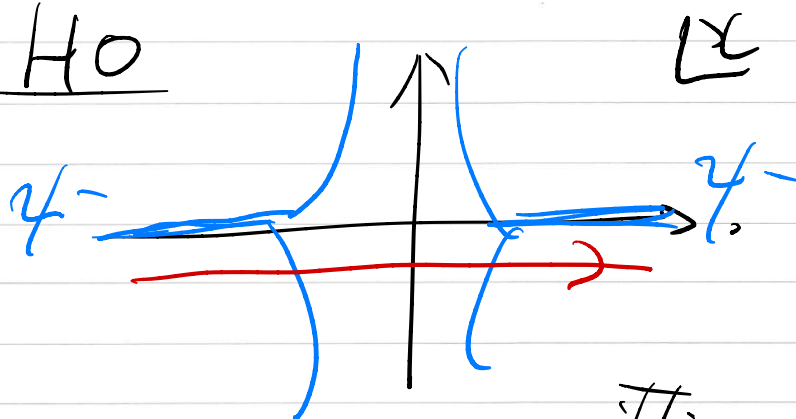
$$\text{in } \text{Im} S \int \sqrt{V-E} = 0$$

* Laplace of x

Bound state

$$x \in (-x, x)$$

H0

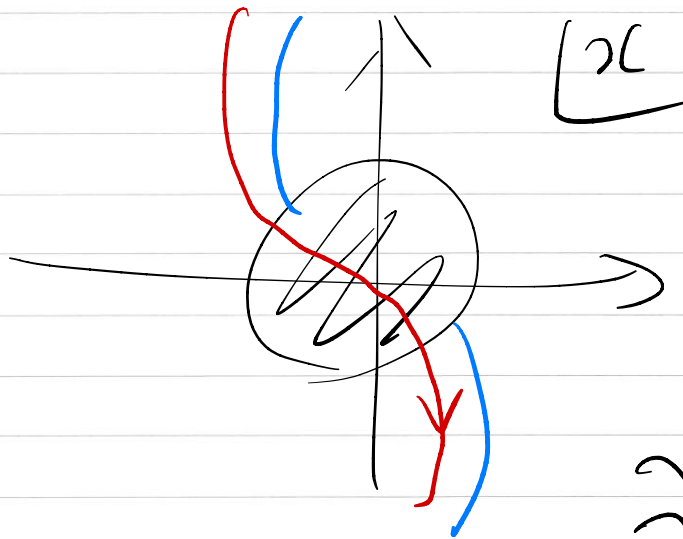


integral
from $x = -\infty$
to $x = \infty$

omitting divergent ψ ,

$$E_n = \hbar \left(n + \frac{1}{2} \right)$$

Resonance



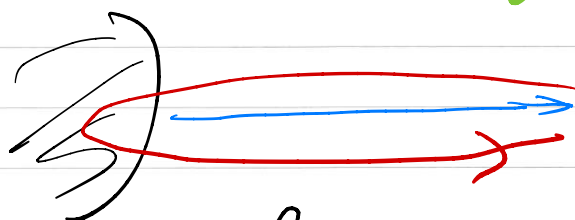
bound state
with diff norm

$\approx$ complex scaling

Reproduce ABC in physics

BH *Miyachi* with appropriate
initial cond.

Andersson '95



Actually

($0M - 0sum$)

$\approx$ ~~x~~

@ Subtleties in non-Hermitian
if there exist
an self-adjoint H_a
and "non-unitary" trans. S

$S H_a S^{-1} = H$ is
well-behaved quantum th.
 $\Rightarrow$ regularization, (thanks to
elliptic regularity)

$$\text{Tr}(D) = \text{Tr}(D e^{-D})$$

$\downarrow$
anomaly

$$\text{Tr}(D e^{-D}) < \infty$$

Drastically,

$$H = \underbrace{\square}_{\text{defective}}$$